

Lab_09

Hadia Beshir

Import Data

```
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

> Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

According to the function there are 85 different candy types in this data set.

> Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

According to the function there is 38 different fruity candy types in this dataset.

We can find the winpercent value for Twix by using its name to access the corresponding row of the dataset. This is because the dataset has each candy name as rownames (recall that we set this when we imported the original CSV file). For example the code for Twix is:

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
candy |>
  filter(row.names(candy)== "Reese's Peanut Butter cup") |>
  select(winpercent)
```

```

              winpercent
Reese's Peanut Butter cup 84.18029
```

The winpercent value for Reese's is 84.18.

> **Q4. What is the winpercent value for “Kit Kat”?**

```
candy |>
  filter(row.names(candy)== "Kit Kat") |>
  select(winpercent)
```

```

              winpercent
Kit Kat      76.7686
```

The winpercent value for Kit Kat is 76.77

> **Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?**

```
candy |>
  filter(row.names(candy)== "Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```

              winpercent
Tootsie Roll Snack Bars 49.6535
```

The winpercent value for Tootsie roll is 49.65

There is a useful `skim()` function in the `skimr` package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

```
summary(candy)
```

chocolate	fruity	caramel	peanutyalmondy
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.4353	Mean :0.4471	Mean :0.1647	Mean :0.1647

3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.0000
nougat	crispedricewafer	hard	bar
Min. :0.00000	Min. :0.00000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.00000	Median :0.00000	Median :0.0000	Median :0.0000
Mean :0.08235	Mean :0.08235	Mean :0.1765	Mean :0.2471
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.00000	Max. :1.00000	Max. :1.0000	Max. :1.0000
pluribus	sugarpercent	pricepercent	winpercent
Min. :0.0000	Min. :0.0110	Min. :0.0110	Min. :22.45
1st Qu.:0.0000	1st Qu.:0.2200	1st Qu.:0.2550	1st Qu.:39.14
Median :1.0000	Median :0.4650	Median :0.4650	Median :47.83
Mean :0.5176	Mean :0.4786	Mean :0.4689	Mean :50.32
3rd Qu.:1.0000	3rd Qu.:0.7320	3rd Qu.:0.6510	3rd Qu.:59.86
Max. :1.0000	Max. :0.9880	Max. :0.9760	Max. :84.18

>Q7. What do you think a zero and one represent for the candy\$chocolate column?

```
table(candy$chocolate)
```

```
0 1
48 37
```

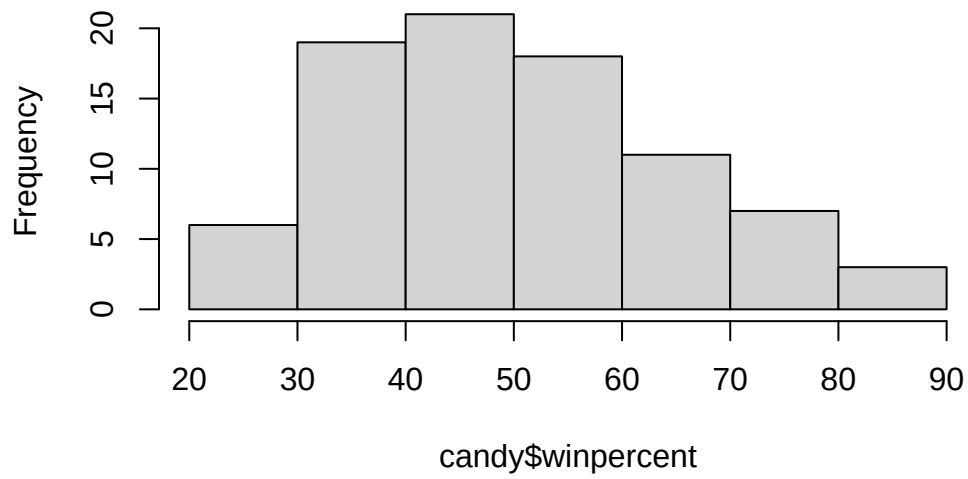
A zero in the candy chocolate column shows that there are that number of candies that do not contain chocolate (false), and a 1 means that there are that amount of candies that do contain chocolate. In this case, 48 candies don't have chocolate and 37 do.

> Q8. Plot a histogram of winpercent values using both base R and ggplot2.

Plot using R

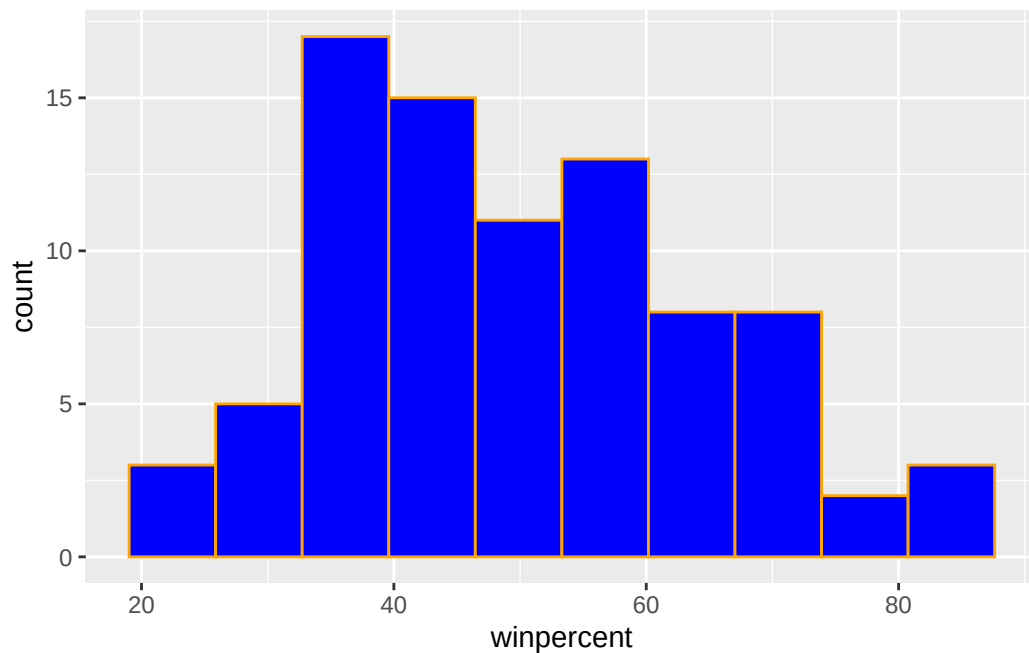
```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



Plot using ggplot2

```
library(ggplot2)
ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=10, fill="blue", col="orange")
```



> **Q9. Is the distribution of winpercent values symmetrical?**

It is clear that the distribution is slightly skewed to the right. There are two peak areas in the data. Despite the two peaks the data is relatively packed in the middle.

> **Q10. Is the center of the distribution above or below 50%?**

```
median(candy$winpercent)
```

```
[1] 47.82975
```

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

According to the functions ran above, the center of distribution (better explained by the median) is below 50%.

> **Q11. On average is chocolate candy higher or lower ranked than fruit candy?**

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

According to the data, chocolate has a higher average than fruity does. This suggests that chocolate is higher ranked.

> **Q12. Is this difference statistically significant?**

```
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The p-value, as provided above, is 2.871e-08. If the p-value is less than 0.05, then the result is significant. Since the p-value is far below 0.05, the results have high statistical significance.

> **Q13. What are the five least liked candy types in this set?**

```
head(candy[order(candy$winpercent), ], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped rice	wafer	hard bar	pluribus	sugar	percent	price	percent
Nik L Nip		0	0	0	1	0.197		0.976
Boston Baked Beans		0	0	0	1	0.313		0.511
Chiclets		0	0	0	1	0.046		0.325

Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
candy %>% arrange(winpercent) %>% head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip		0	0	0		1		0.197		0.976
Boston Baked Beans		0	0	0		1		0.313		0.511
Chiclets		0	0	0		1		0.046		0.325
Super Bubble		0	0	0		0		0.162		0.116
Jawbusters		0	1	0		1		0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The five least liked candy types in this set: 1. Nik L Nip
 2. Boston Baked Beans
 3. Chiclets 4. Super Bubble 5. Jawbusters

> Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE), ], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0

Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1
	crisped	ricewafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0.720
Reese's Miniatures		0	0	0	0.034
Twix		1	0	1	0.546
Kit Kat		1	0	1	0.313
Snickers		0	0	1	0.546
	pricepercent	winpercent			
Reese's Peanut Butter cup	0.651	84.18029			
Reese's Miniatures	0.279	81.86626			
Twix	0.906	81.64291			
Kit Kat	0.511	76.76860			
Snickers	0.651	76.67378			

```
candy %>% arrange(desc(winpercent)) %>% head(5)
```

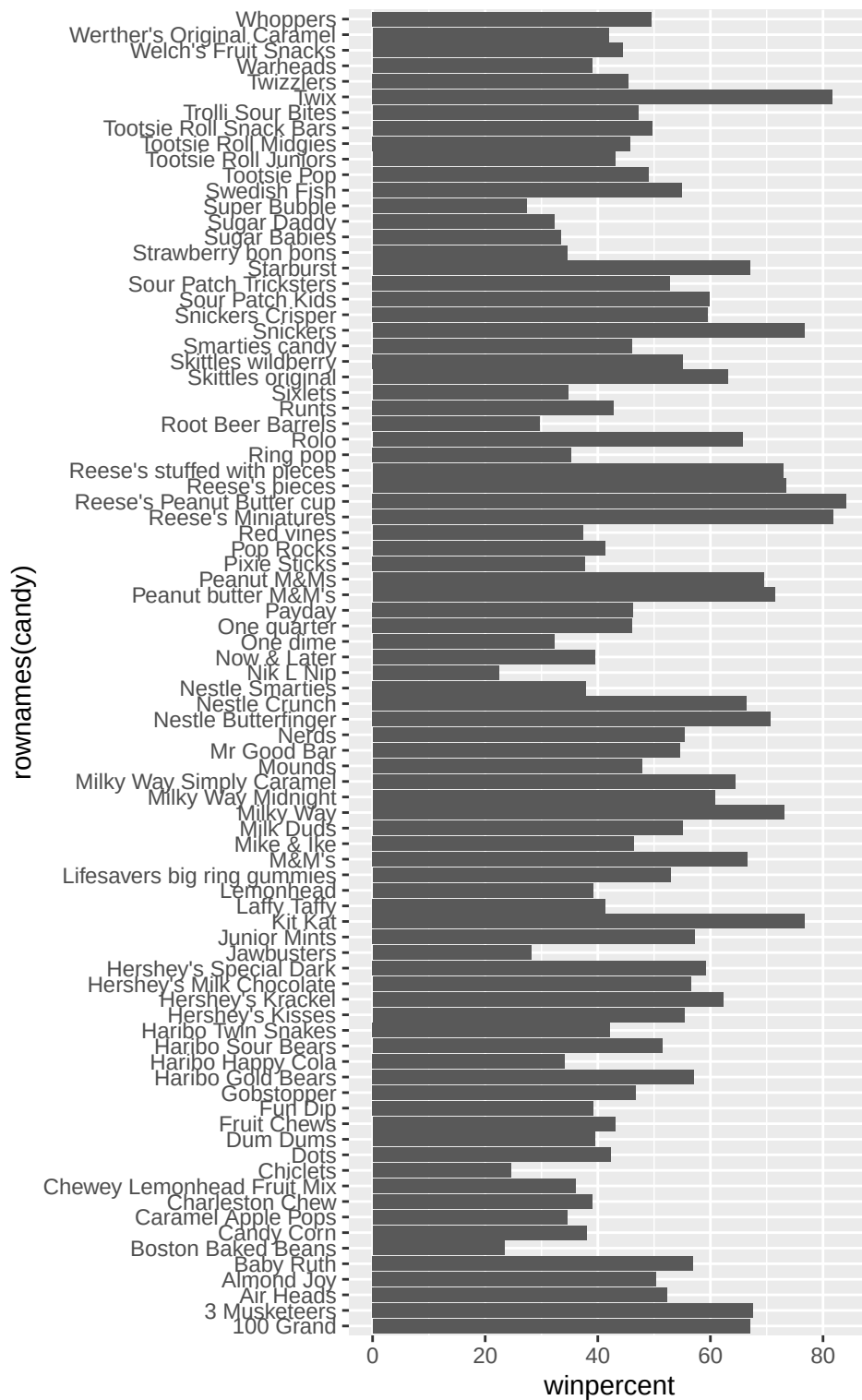
	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1
	crisped	ricewafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0.720
Reese's Miniatures		0	0	0	0.034
Twix		1	0	1	0.546
Kit Kat		1	0	1	0.313
Snickers		0	0	1	0.546
	pricepercent	winpercent			
Reese's Peanut Butter cup	0.651	84.18029			
Reese's Miniatures	0.279	81.86626			
Twix	0.906	81.64291			
Kit Kat	0.511	76.76860			
Snickers	0.651	76.67378			

The top 5 all time favorite candy types out of this set are: 1. Reese's Peanut Butter cup
 2. Reese's Miniatures
 3. Twix 4. Kit Kat 5. Snickers

> Q15. Make a first barplot of candy ranking based on winpercent values.

```
library(ggplot2)

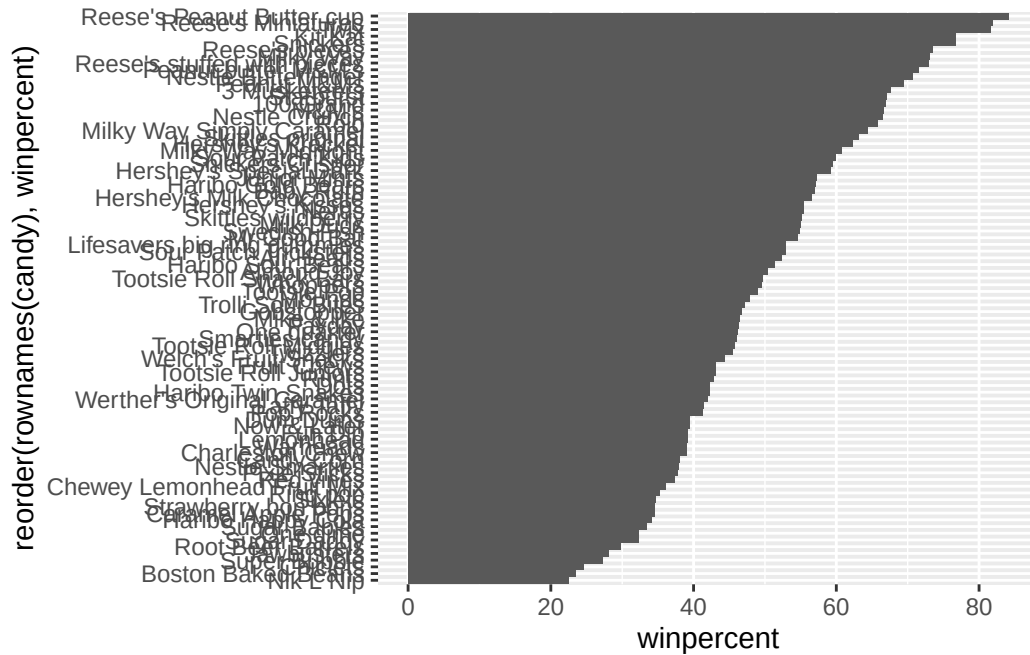
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



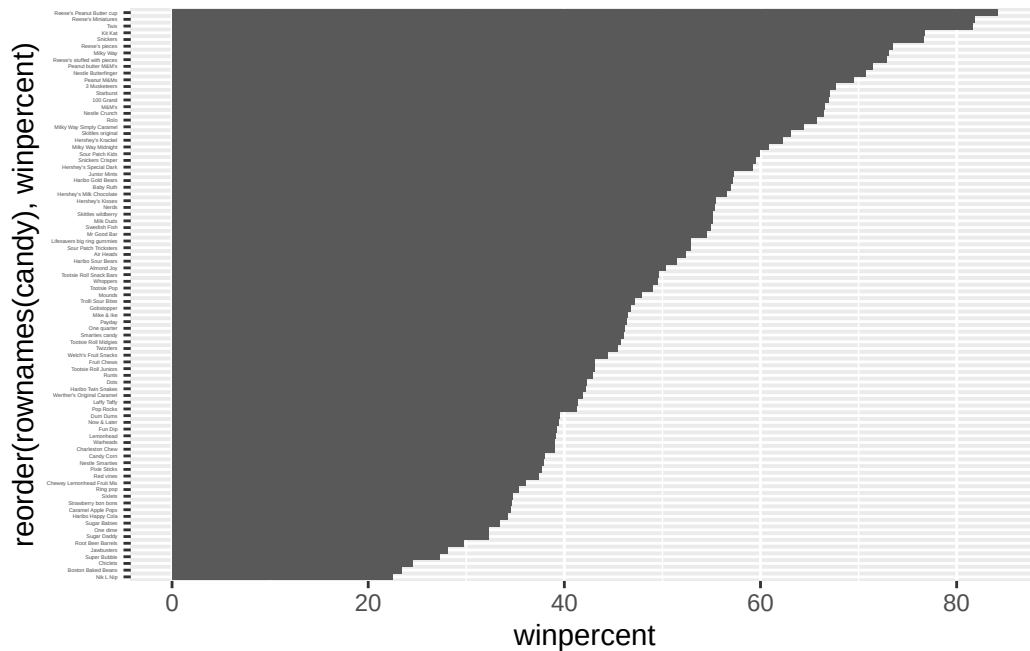
> Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```

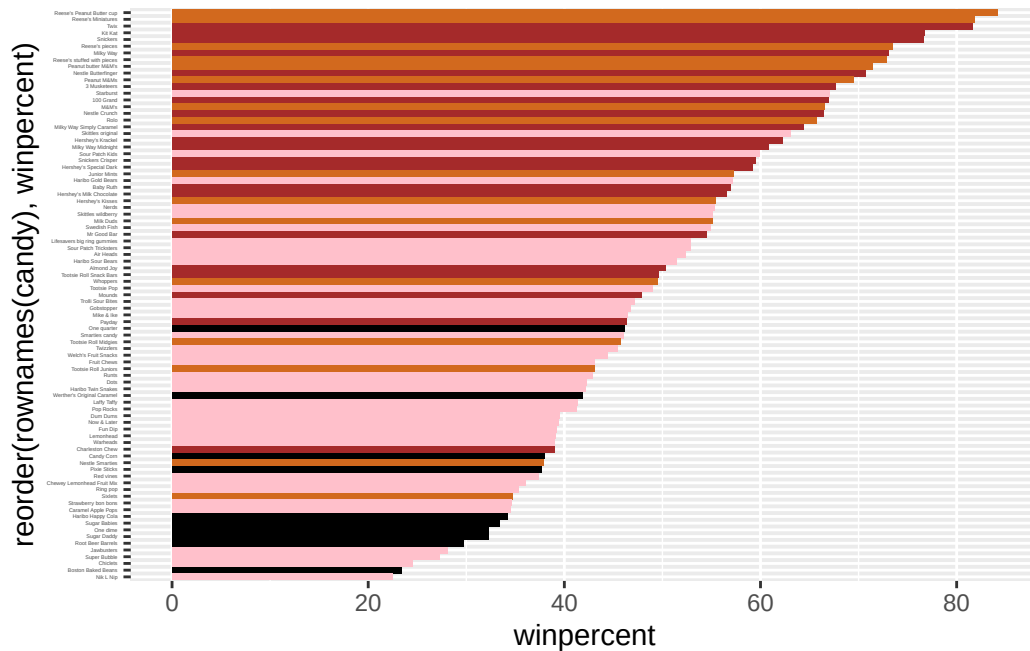


```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col() +
  theme(axis.text.y = element_text(size = 2))
```



Let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  theme(axis.text.y = element_text(size = 2))
```



> Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Reeses peanut Butter cup.

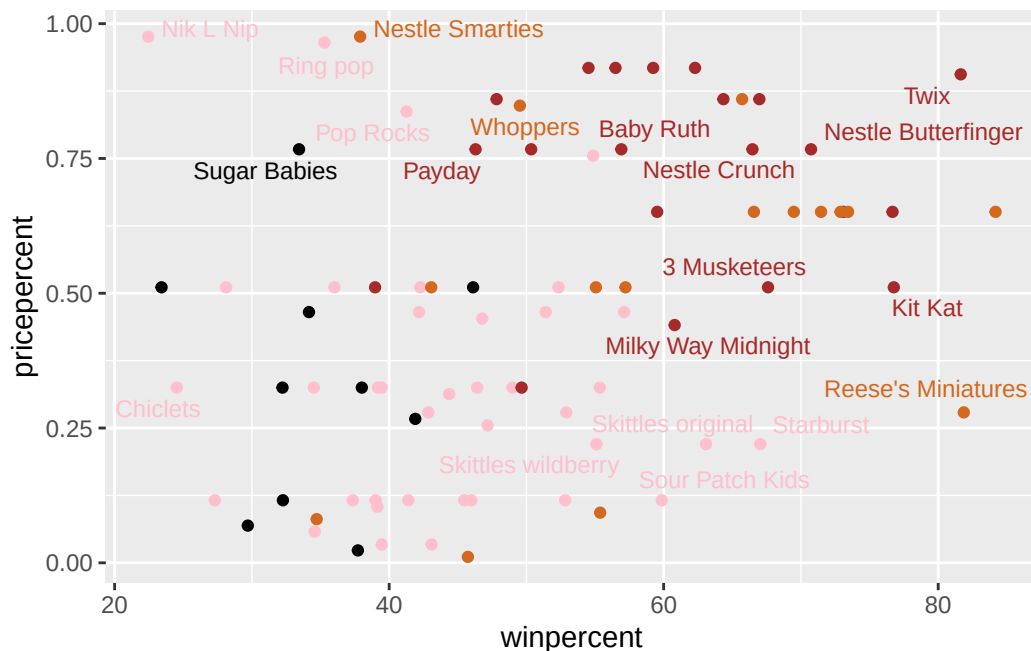
> Q18. What is the best ranked fruity candy?

The best ranked frity candy is Starburst.

Taking a look at pricepercent

To this plot we will add text labels so we can more easily identify a given candy. There is a regular `geom_label()` that comes with `ggplot2`. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the `geom_text_repel()` function from the `ggrepel` package.

```
library(ggrepel)
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



>Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
head(candy[order(candy$pricepercent), c("pricepercent", "winpercent")], n=5)
```

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675
Pixie Sticks	0.023	37.72234
Dum Dums	0.034	39.46056
Fruit Chews	0.034	43.08892
Strawberry bon bons	0.058	34.57899

Tootsie Roll Midgies is the cheapest candy and has the most bang for your buck.

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534

Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

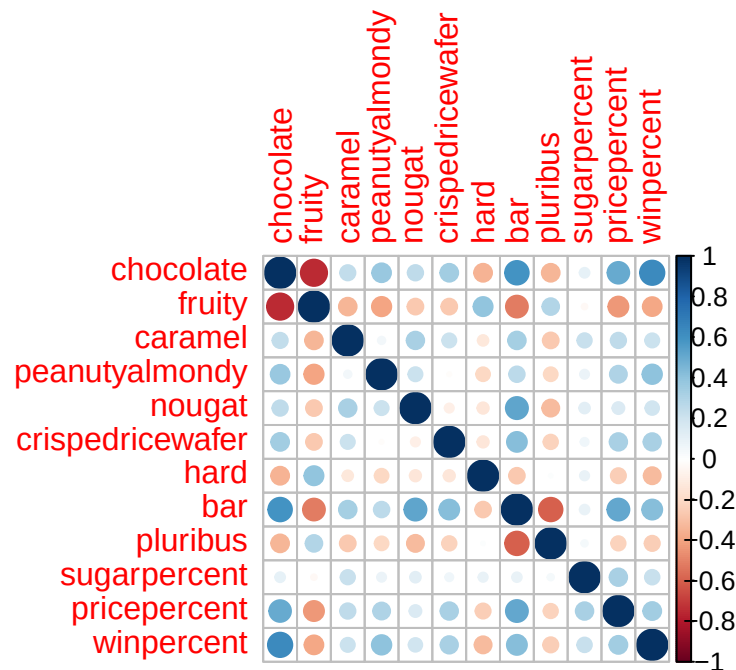
The most expensive candy and the least popular are: 1. Nik L Nip
 2. Nestle Smarties
 3. Ring pop 4. Hershey's Krackel
 5. Hershey's Milk Chocolate

Exploring the correlation structure

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)? The two most aniti correlated variables is fruity and chocolate.

> Q23. Similarly, what two variables are most positively correlated?

The two most positively correlated variables are winipercnt and chocolate.

Principal Component Analysis

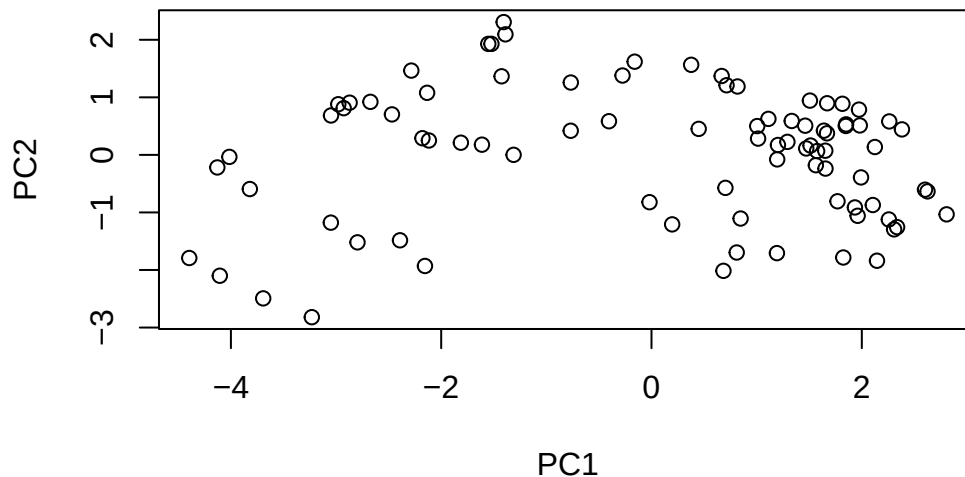
```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

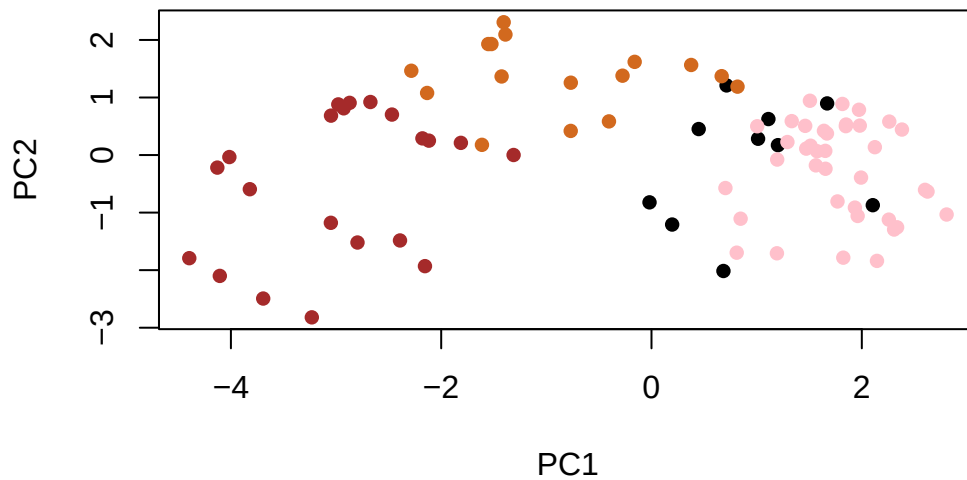
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



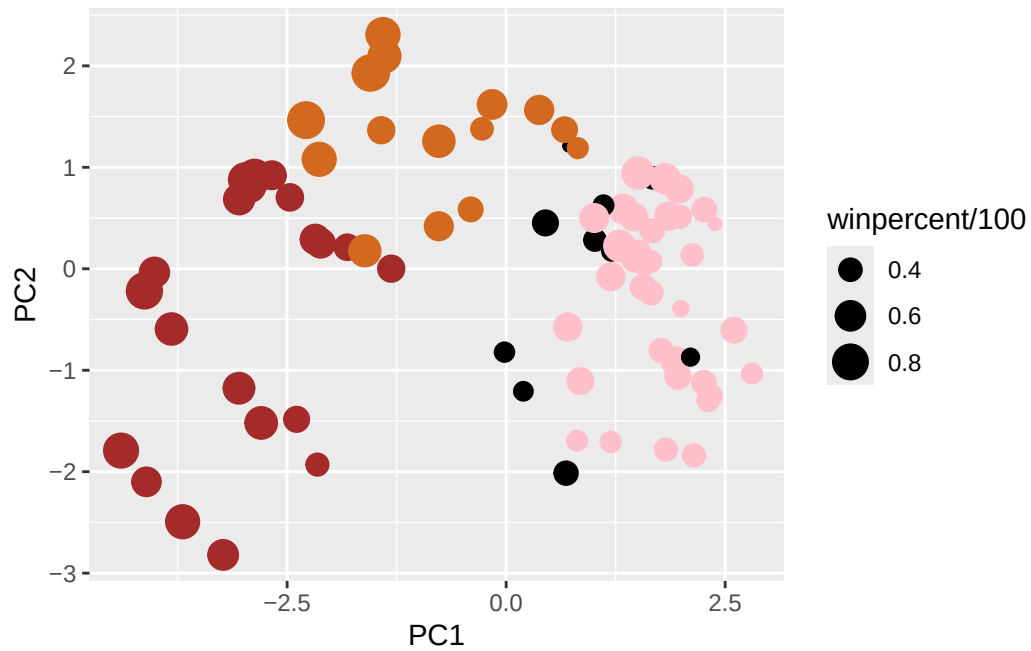
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



We can make a much nicer plot with the ggplot2 package but it is important to note that ggplot works best when you supply an input data.frame that includes a separate column for each of the aesthetics you would like displayed in your final plot.

```
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
p
```

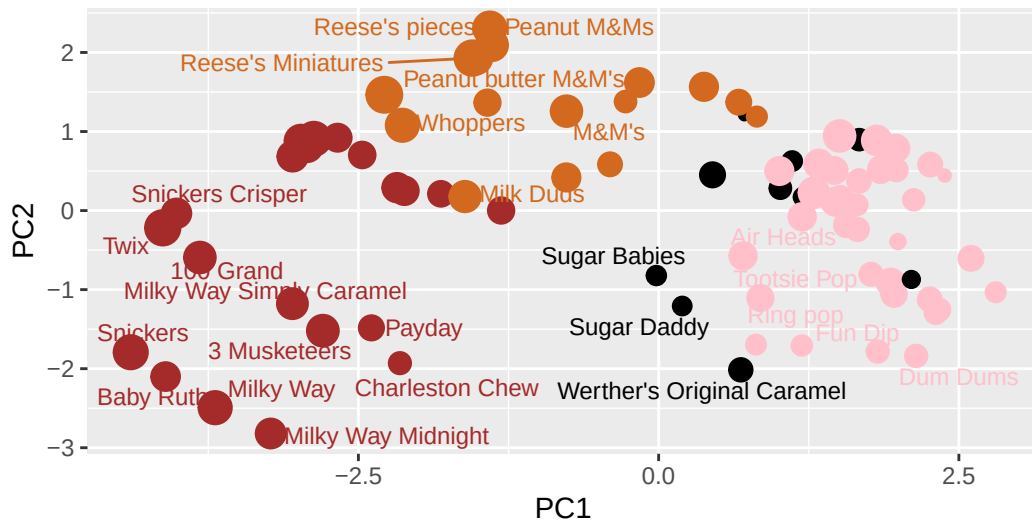


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

If you want to see more candy labels you can change the `max.overlaps` value to allow more overlapping labels or pass the ggplot object `p` to `plotly` like so to generate an interactive plot that you can mouse over to see labels:

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

`last_plot`

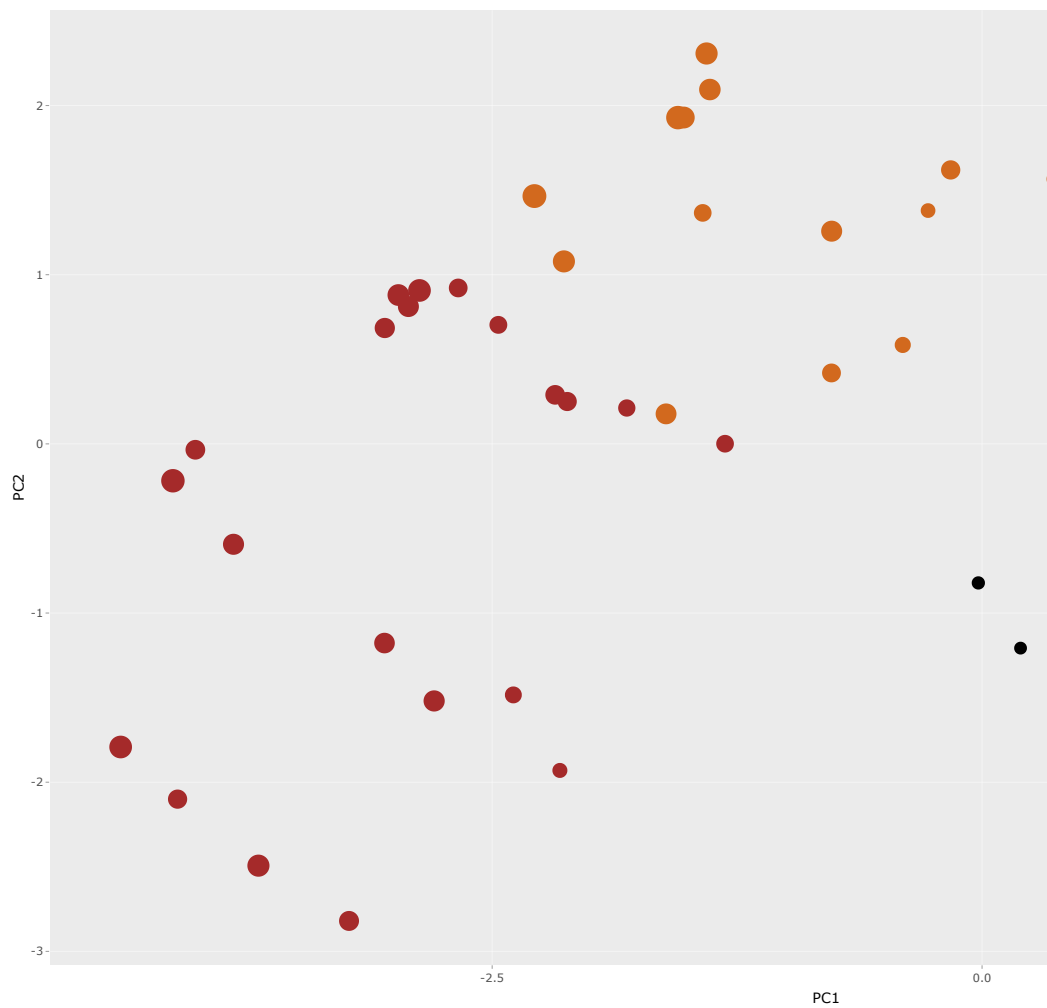
The following object is masked from 'package:stats':

`filter`

The following object is masked from 'package:graphics':

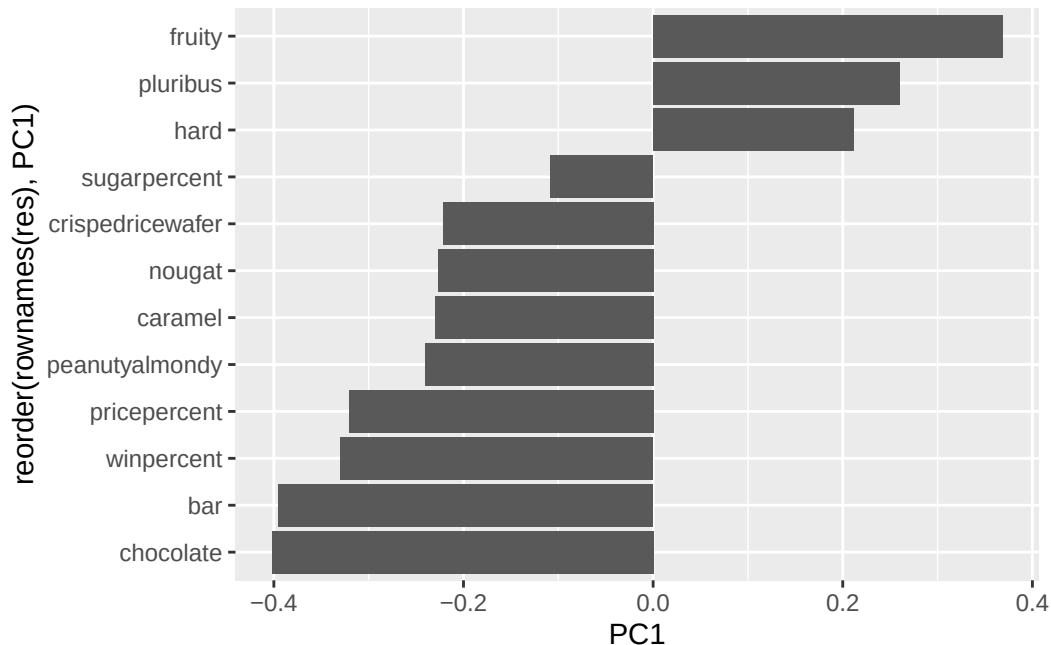
`layout`

```
ggplotly(p)
```



Let's finish by taking a quick look at PCA our loadings. Do these make sense to you? Notice the opposite effects of chocolate and fruity and the similar effects of chocolate and bar (i.e. we already know they are correlated).

```
res <- as.data.frame(pca$rotation)
ggplot(res) +
  aes(PC1, reorder(rownames(res), PC1)) +
  geom_col()
```



>Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The variables that have the most positive direction and point to the right are fruity, pluribus, and hard. This correlation has been seen in the plots above, especially the anti correlation plot where fruity area was on the right while chocolate was on the left. I would say this does make sense because fruity candies including gummies, sourpatch, skittles, are all coming in small packs and are slightly harder and more chewy.

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Based on the plots we generated so far and my understanding of the test is that people like chocolate (mainly chocolates that have biscuits or peanut flavor in it) the most and are willing to spend their money on it. Many are okay with buying cheaper fruity candies, however, as soon as the price goes up, people are not likely to buy.